8	**(a)(i)**	Activity A of a radioactive substance is defined as the number of disintegrations per unit time.	B1
	(a)(ii)	use of $\lambda = (\ln 2)/t_{½}$	C1
		shows that years have been converted to seconds	
		$\lambda = 2.5 \times 10^{-10}\ s^{-1}$	A1
	(a)(iii)	$N = A/\lambda$	
		$= 0.63 \times 10^{12}/2.5 \times 10^{-10}$ per g (must use value given in the table)	C1
		$= 0.63 \times 10^{12}/2.5 \times 10^{-10} \times 0.16$	C1
		$= 4.0 \times 10^{20}$	
		1 mark awarded if use	A0
		$N = (0.16/238) \times 6.02 \times 10^{23} = 4.0 \times 10^{20}$	
	(b) (i)	Energy per unit time = activity x energy per decay	C1
		Energy per sec = $(0.63 \times 10^{12} \times 0.16) \times 5600 \times 10^{3} \times 1.60 \times 10^{-19}$	C1
		= 0.090 J	A1
	(b) (ii)	$e = 7.5 \times 10^{-4}/0.090 \times 100\%$	C1
		= 0.83%	A1
	(c)	Since Power output is proportional to Activity,	
		$P = P_o e^{-\lambda t}$	C1
		$0.60/0.75 = e^{-\lambda t}$	A1
		$t = 8.9 \times 10^{8}$ s	
	(d)	use $E = mc\Delta\theta$	C1
		$0.68 \times 10^{-3} \times 12 \times 3600 = 6.7 \times 10^{-3} \times 410 \times \Delta\theta$	A1
		$\Delta\theta = 11\ ^{o}C$	
	(e)	-A **slow neutron may cause nuclear fission of Pu-238 and trigger a chain reaction**, which will cause a high amount of energy to be released in a short time. This is undesirable for the pace-maker.	B1
		-It is important to investigate the decay chain to **eliminate the possibility of alpha or gamma emitters in the decay series** of the parent nucleus	B1
	(f)	-**Reject alpha emitters because it can lead to neutron production** and select beta emitters	
		And anyone:	
		-**reject Strontium-90 as it is most energetic** of the 3 beta emitters, since it **could produce gamma radiation**	B1
		OR	
		-**reject Hydrogen-3** due to its **short half-life** (as compared to average human life span), therefore **need constant replacement** (Any one correct reason)	
		And	
		-**select Nickel-63** as energy released per decay is suitable/ it has an appropriate half-life (as compared to average human lifespan)	B1